

SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – PROBLEM STATEMENT

Course Name	Artificial Intelligence	Course Code	CSA17
Annexure	A – Resolution Algorithm Implementation	Total Marks	30
Course Outcome	CO3 (BL4)	Weightage	20%

Course Outcome Covered (CO3): Analyze knowledge representation and reasoning using propositional logic, First-Order Logic (FOL), inference, resolution and chaining methods, and knowledge representation to perform logical reasoning for drawing valid conclusions and decision making.

Overall Objective: For each scenario below, formalize the given facts and rules in propositional logic, convert every statement to Conjunctive Normal Form (CNF), negate the goal, and apply the Resolution Algorithm step-by-step until the Empty Clause (NIL) is derived, thereby proving the goal follows logically from the knowledge base.

QUESTIONS

Question 1 – Rain and Wet Ground

Knowledge Base:

- 1 If it rains, then the ground becomes wet.
- 2 It is raining.

Goal: Prove that the ground is wet using the Resolution Algorithm.

Tasks:

- a Represent the statements in propositional logic.
- b Convert all statements into CNF.
- c Negate the goal.
- d Apply the Resolution Algorithm step by step.
- e Show the Empty Clause (NIL) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base:

- 1 If a student submits the assignment, then the student receives internal marks.
- 2 Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks using the Resolution Algorithm.

Tasks:

- a Write the propositional logic expressions.
- b Convert the premises into CNF.

- c Negate the goal.
 - d Apply the Resolution Algorithm.
 - e Conclude whether the goal is proved.
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Question 3 – Library Membership

Knowledge Base:

- 1 If a person is a library member, then the person can borrow books.
- 2 Priya is a library member.

Goal: Prove that Priya can borrow books using the Resolution Algorithm.

Tasks:

- a Represent the knowledge base using propositional logic.
 - b Convert each statement into CNF.
 - c Negate the goal.
 - d Perform resolution until no new clauses can be generated.
 - e State the final conclusion.
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Question 4 – Placement Eligibility

Knowledge Base:

- 1 If a student clears the aptitude test, then the student is eligible for placement.
- 2 Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement using the Resolution Algorithm.

Tasks:

- a Convert the statements into propositional logic.
 - b Write the CNF clauses.
 - c Negate the goal.
 - d Apply the Resolution Algorithm step by step.
 - e Draw the resolution tree and conclude.
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Question 5 – Access Control System

Knowledge Base:

- 1 If a user enters the correct password, then the user is authenticated.
- 2 If a user is authenticated, then the user is granted access.
- 3 The user entered the correct password.

Goal: Prove that the user is granted access using the Resolution Algorithm.

Tasks:

- a Represent the knowledge base in propositional logic.
 - b Convert all implications into CNF.
 - c Negate the goal.
 - d Apply the Resolution Algorithm to derive the Empty Clause (NIL).
 - e Explain each resolution step and write the final conclusion.
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